

Heaps / Priority Queue

Complete binary tree with heap order.

Key concepts

- Max-heap: parent $\geq$ children; min-heap opposite.
- Array-backed: children at $2i+1$, $2i+2$.
- insert/extract in $O(\log n)$.

Key idea / example

Priority queue returns the highest-priority item first.

Today's tasks (1.5h)

- ☐ Implement a binary heap
- ☐ Do insert & extract-max
- ☐ Use it as a priority queue

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)